

In supervised learning, the data are pairs (x_i, y_i) , where $x_i \in \mathcal{X}$ is the input and $y_i \in \mathcal{Y}$ is the label or targets.

We assume that the data are sampled from some unknown distribution $\mathcal{D}$: $(x_i, y_i) \sim \mathcal{D}$ for $i = 1, 2, \dots, n$.

A model is a function $f: \mathcal{X} \rightarrow \mathcal{Y}$. Given an input x , it outputs a prediction $f(\vec{x})$. For example in regression, $f(\vec{x}) \in \mathbb{R}$.

In binary classification, $f(\vec{x}) \in \{0, 1\}$. In neural networks,

f might be a network with parameters w is denoted by

$f_w(\vec{x})$. The set of all predictors we allow is called the

hypothesis class, denoted by $\mathcal{H}$. For example,

$$\mathcal{H} = \{ \vec{x} \mapsto \vec{w}^T \vec{x} : \|\vec{w}\|_2^2 \leq B \},$$

or for neural networks:

$$\mathcal{H} = \{ \text{all neural networks with fixed architecture} \}$$

Population Risk:

The quantity we actually care about is the expected loss

$$R(f) = \mathbb{E}_{(x,y) \sim \mathcal{D}} [\ell(f(x), y)] .$$

This is called the **population risk**. The problem is that $\mathcal{D}$ is unknown, so we cannot compute $R(f)$ exactly.

Empirical Risk:

What we can compute is the empirical risk:

$$\hat{R}_s(f) = \frac{1}{n} \sum_{i=1}^n \ell(f(x_i), y_i)$$

Suppose, we choose a function f from $\mathcal{H}$ in advance, i.e. before seeing data. Furthermore, assume that $0 \leq \ell(f(x), y) \leq 1$.

Define the random variables $\mathcal{L}_i = \ell(f(x_i), y_i)$. Since (x_i, y_i) are i.i.d., the losses are also i.i.d. Therefore, their expectation is

$$\mathbb{E}[\mathcal{L}_i] = R(f).$$

So $\hat{R}_s(f)$ is just the sample average of random variables whose mean is $R(f)$. Then by Hoeffding's inequality,

$$\Pr\{|R(f) - \hat{R}_s(f)| \geq \varepsilon\} \leq 2e^{-2n\varepsilon^2}$$

Hoeffding's Inequality:

Let $X_1, X_2, \dots, X_n$ be independent random variables such that

$\Pr \{a_i \leq x_i \leq b_i\} = 1$. Consider $S_n = X_1 + \dots + X_n$. Then,

Hoeffding's theorem states that, for all $t > 0$,

$$\Pr \{|S_n - \mathbb{E}[S_n]| \geq t\} \leq 2 \exp \left(- \frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2} \right).$$

$$\Pr \{|R(f) - \hat{R}_S(f)| \geq \epsilon\} \leq 2e^{-2n\epsilon^2}$$

$$\Rightarrow \Pr \{|\hat{R}_S(f) - R(f)| < \epsilon\} \geq 1 - 2e^{-2n\epsilon^2}$$

Let $\delta := 2e^{-2n\epsilon^2}$. Then $\epsilon = \sqrt{\frac{\log(2/\delta)}{2n}}$. This means

that for a fixed hypothesis f from $\mathcal{H}$, with probability at least $1 - \delta$,

$$|R(f) - \hat{R}_S(f)| \leq \sqrt{\frac{\log(2/\delta)}{2n}}$$

Q: We derived a generalization bound for a hypothesis f fixed before seeing data. Can we derive a generalization bound for a hypothesis f that is learned by an algorithm A , which chooses a hypothesis on $\mathcal{H}$?

A: Suppose algorithm A takes training data $S := \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$ and returns a model $\hat{f}$. Since $\hat{f}$ is a function of

5, $\ell(\hat{f}(x_i), y_i)$'s are not i.i.d anymore. Therefore, the previous analysis does not hold anymore. Furthermore, the algorithm $\mathcal{A}$ does not have to be optimal for $\mathcal{H}$ always, i.e.

$$\mathcal{A}(S) = \underset{f \in \mathcal{H}}{\operatorname{argmin}} \hat{R}_S(f).$$

Hence, the learning algorithm may choose any function in $\mathcal{H}$ after seeing data. Therefore, we want a guarantee that holds uniformly over $\mathcal{H}$.

Suppose $\exists f \in \mathcal{H} : |R(f) - \hat{R}_S(f)| > \epsilon$. This implies

$\bigcup_{f \in \mathcal{H}} |R(f) - \hat{R}_S(f)| > \epsilon$. Therefore, by using the union bound,

$$\mathbb{P}_n \left\{ \exists f \in \mathcal{H} : |R(f) - \hat{R}_S(f)| > \epsilon \right\} \leq \sum_{f \in \mathcal{H}} \mathbb{P}_n \left\{ |R(f) - \hat{R}_S(f)| > \epsilon \right\}.$$

Each term is bounded by Hoeffding's inequality

$$\mathbb{P}_n \left\{ |R(f) - \hat{R}_S(f)| > \epsilon \right\} \leq 2e^{-2n\epsilon^2}$$

$\Rightarrow$

$$\mathbb{P}_n \left\{ \exists f \in \mathcal{H} : |R(f) - \hat{R}_S(f)| > \epsilon \right\} \leq 2|\mathcal{H}|e^{-2n\epsilon^2}$$

$$1 - \mathbb{P}_n \left\{ \exists f \in \mathcal{H} : |R(f) - \hat{R}_S(f)| > \epsilon \right\} \geq 1 - 2|\mathcal{H}|e^{-2n\epsilon^2}.$$

$$\Leftrightarrow \mathbb{P}_n \left\{ \forall f \in \mathcal{H} : |R(f) - \hat{R}_S(f)| \leq \epsilon \right\} \geq 1 - \delta, \text{ where}$$

$\delta = 2|\mathcal{H}|e^{-2n\epsilon^2}$. This means that with probability at least $1-\delta$,

$$|R(f) - \hat{R}_s(f)| \leq \sqrt{\frac{\log(\frac{2|\mathcal{H}|}{\delta})}{2n}} \quad \forall f \in \mathcal{H}.$$

Excess Risk:

Empirical risk minimization chooses a function with minimum training loss:

$$\hat{f} \in \operatorname{argmin}_{f \in \mathcal{H}} \hat{R}_s(f)$$

However, we care about population risk, not empirical

risk:

$$f_{\mathcal{H}}^* \in \operatorname{argmin}_{f \in \mathcal{H}} R(f).$$

$f_{\mathcal{H}}^*$ is called **best-in-class predictor**.

The excess risk of $\hat{f}$ relative to $\mathcal{H}$ is

$$R(\hat{f}) - R(f_{\mathcal{H}}^*).$$

If the excess risk is small, then ERM learned the best function in $\mathcal{H}$.

Theorem: If $\sup_{f \in \mathcal{H}} |R(f) - \hat{R}_s(f)| \leq \epsilon$, then

$$R(\hat{f}) - R(f_{\mathcal{H}}^*) \leq 2\epsilon$$

Proof:

$$\begin{aligned}
 & \underbrace{R(\hat{f}) - \hat{R}_S(\hat{f})}_{\leq \epsilon} + \underbrace{\hat{R}_S(\hat{f}) - \hat{R}_S(f_{\mathcal{H}}^*)}_{\leq 0} + \underbrace{\hat{R}_S(f_{\mathcal{H}}^*) - R(f_{\mathcal{H}}^*)}_{\leq \epsilon} \\
 \Rightarrow & R(\hat{f}) - R(f_{\mathcal{H}}^*) \leq 2 \sup_{f \in \mathcal{H}} |R(f) - \hat{R}_S(f)| \leq 2\epsilon.
 \end{aligned}$$

Sample Complexity:

Sample complexity is the number of data points required to make the generalization error small with high probability. For example, in the finite hypothesis class case, we had

$$R(\hat{f}) - R(f_{\mathcal{H}}^*) \leq 2 \sqrt{\frac{\log(2|\mathcal{H}|/\delta)}{2n}}$$

If we want this to be at most ϵ , we need roughly,

$$\begin{aligned}
 n & \gtrsim \frac{\log(|\mathcal{H}|) + \log(1/\delta)}{\epsilon^2} \\
 n & = O\left(\frac{\log(|\mathcal{H}|) + \log(1/\delta)}{\epsilon^2}\right)
 \end{aligned}$$

Remark: To learn well, you need more samples when the hypothesis class is larger, when you want smaller error ϵ , or when you want higher confidence.